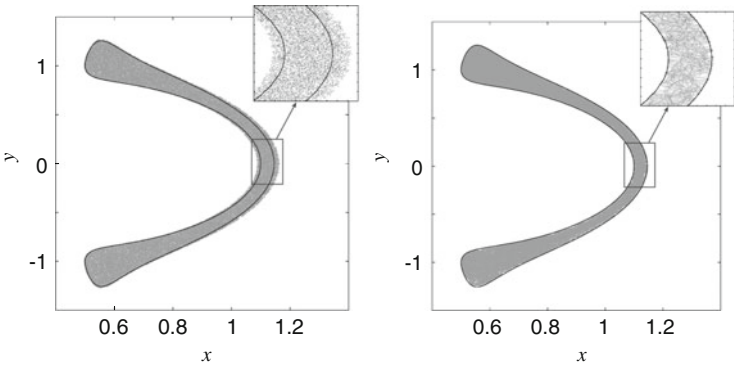


Fig. 5.8 Long trajectories of the isosceles 3-body problem computed using two different timesteps, $h = 0.07$ (*left*) and $h = 0.01$ (*right*). The discrete values have been used to draw *small dots* which appear to fill in a region bounded by a level curve of the potential energy, with some error at the boundary that decreases with stepsize h

have plotted two long numerical solutions (100,000 timesteps) in their xy projection for two different stepsizes $h = 0.07$ and $h = 0.01$, started from a point with energy $\mathrm{E} = -0.5$ which is (a) admissible and (b) connected and (c) bounded. The dark curve represents the level curve of the potential energy for $U = -0.5$ while the numerical steps are shown in gray. As we can see in this figure, the points appear to shade the region completely, however spilling over a bit from the interior of the curve. This overspill is due to the numerical error, as it is decreased by a decrease in the stepsize.

As we have seen in the Chap. 3, when a symplectic method is applied to a molecular dynamics problem it induces a perturbed Hamiltonian (energy) function. For the Verlet scheme the modified Hamiltonian is

$$\tilde{H}_h = H + \frac{h^2}{24}\left(2\boldsymbol{p}^T\boldsymbol{M}^{-1}U''(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} - \nabla U(\boldsymbol{q})\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\right) + \mathcal{O}(h^4).$$

Actually, as explained previously, we must interpret this carefully: a finite truncation of this expansion may characterize the discrete trajectories observed in practice, but the series expansion itself does not typically converge. Assuming such a perturbative expansion is valid, the issue of ergodicity of the numerical method can be related to the corresponding issue for the dynamical system characterized by $\tilde{H}_h$.

A result in [338] provides a formula to calculate the perturbation of the microcanonical average induced by the Hamiltonian perturbation. This result has been recast in the more convenient form in [14]:

Theorem 5.1 *Let $H(\boldsymbol{q},\boldsymbol{p})$ and $\tilde{H}(\boldsymbol{q},\boldsymbol{p}) = H(\boldsymbol{q},\boldsymbol{p}) + \varepsilon\eta(\boldsymbol{q},\boldsymbol{p})$ be two Hamiltonian functions. Suppose the energy surface $\Sigma_{\mathrm{E}} = \{(\boldsymbol{q},\boldsymbol{p})|H(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}\}$ to be compact and define the perturbed energy surface by $\tilde{\Sigma}_{\mathrm{E}} = \{(\boldsymbol{q},\boldsymbol{p})|\tilde{H}(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}\}$, also assumed compact.*

Suppose that $\boldsymbol{w}$ *is an arbitrary smooth vector field such that* $\boldsymbol{w} \cdot \nabla H \neq 0$ *on* Σ_{E}. *Then the perturbation of the microcanonical average of a smooth function* g *is given by*

$$\mathrm{Av}_{H=\mathrm{E}}(g) - \mathrm{Av}_{\tilde{H}=\mathrm{E}}(g) = \mathrm{Av}_{\tilde{H}=\mathrm{E}}(\nabla \cdot g\boldsymbol{u}) - \mathrm{Av}_{\tilde{H}=\mathrm{E}}(g)\mathrm{Av}_{\tilde{H}=\mathrm{E}}(\nabla \cdot \boldsymbol{u}) + \mathcal{O}(\varepsilon^2),$$

where

$$\boldsymbol{u} = \varepsilon \frac{\eta}{\boldsymbol{w} \cdot \nabla H} \boldsymbol{w}.$$

The point of this result is that it allows all computations to be done in the terms of $\tilde{H}_h$ which is the system we can study using the numerical trajectories (assuming, of course, ergodicity). In other words we can correct an average computed using a symplectic numerical method for the effect of its truncation error. A natural choice of the function $\boldsymbol{w}$ that appears in the theorem is $\boldsymbol{w} = \nabla H$. In [14] this result is applied to the Verlet method, providing an accurate calculation of the averaging error along a numerical trajectory.

The interpretation of a dynamical average (e.g. an autocorrelation function) as a stationary average means that we can apply the perturbation result to such computations as well. For more discussion of numerical averages and computation of correlation functions, see [338].

Exercises

1. For the harmonic oscillator, calculate, for any function $g = g(q, p)$, the action of the Liouvillian $\mathcal{L}_f g$. Also calculate $\exp(t\mathcal{L}_f)g$.
2. Recall from Chap. 3 that a splitting method defines a mapping via a composition of the flow maps associated to additive decomposition of a vector field, i.e. for $\boldsymbol{f} = \boldsymbol{f}_1 + \boldsymbol{f}_2$, then

$$\exp(h\mathcal{L}_f) \approx \exp(h\mathcal{L}_{f_1}) \exp(h\mathcal{L}_{f_2}).$$

   Let ρ be a phase space density. How is the density evolved under such a splitting method?
3. Let $H = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q})$ be a Hamiltonian for N particles, where U is a sum of distance potentials only.
   a. Show by direct calculation that $\mathcal{L}_{\boldsymbol{J}\nabla H} P_x = 0$ where $P_x = \sum_{i=1}^{\mathrm{N}} p_{i,x}$ is the first component of the total momentum.
   b. Show that $\mathcal{L}_{\boldsymbol{J}\nabla H} H = 0$.
   c. Calculate $\mathcal{L}_{\boldsymbol{J}\nabla H} U$.
   d. Calculate $\mathcal{L}^2_{\boldsymbol{J}\nabla H} U$.